

Akash Gogate

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EDUCATION

University of Wisconsin at Madison

Graduating: May 2027

Bachelor of Science in Computer Science • Dean's List Fall 2024, Spring 2025, Fall 2025 • 3.9 GPA

Relevant Coursework: Data Structures, Advanced Algorithms, Machine Organization and Programming, Intro to Computer Engineering, Cryptography, Artificial Intelligence, Differential Equations, Operating Systems, Big Data

South Brunswick High School

Relevant Coursework: Computer Science Capstone, Android Application Development, Virtual Reality, Game Development, Web Development

TECHNOLOGIES

Languages: Java, Python, TypeScript, Node.js, R, Rust, JavaScript, C#, C++, SQL, Linux/Bash

Backend & Systems: Kafka, Kubernetes, Docker, MongoDB, SQLite, Redis, SQL, NoSQL, REST APIs, FastAPI, Supabase, Microservices, CI/CD, Event-Driven Architecture, Distributed Systems, System Design

Frontend & Mobile: React.js, Android SDK, Spring Boot, HTML/CSS

Developer Tools: Git, GitHub, GitHub Actions, Linux, Version Control, pytest, Unit Testing, Agile, Cross-Functional Collaboration, Object-Oriented Design

Libraries & AI: NumPy, Pandas, PyTorch, Scikit-learn, LLM APIs, RAG, OpenCV

EXPERIENCE

Leidos

May 2025 - Present

Software Engineer Intern

- Optimized satellite observation scheduling via memoization, state space reduction, and dynamic programming in Python, **reducing constraint violations** while maintaining full mission objective alignment under real-time operational constraints
- Designed Kafka event-driven data pipeline persisting to MongoDB achieving **low-latency, fault-tolerant streaming and full system decoupling** for mission-critical real-time satellite data operations serving downstream consumers
- Deployed containerized test suite on Kubernetes via **GitHub Actions CI/CD** with comprehensive unit coverage across every component, supporting ongoing development velocity

The Kendzierski Lab at the University of Wisconsin-Madison

September 2025 - Present

Student Research Intern

- Designed and built an agentic system architecture for scRNA-seq pipeline optimization, cutting GPU consumption by **300%** through cross-functional collaboration with interdisciplinary researchers and clinicians across a multi-library Python/R environment
- Engineered TransferAgent module automating citation mapping, literature synthesis, and cross-paper validity scoring for an interdisciplinary research pipeline; reduced review time by **~60%**; GitHub Actions CI/CD managed reproducible pipeline deployment
- Collaborated with clinicians and multidisciplinary researchers to quantify pipeline efficiency gains and validate real-world clinical impact on glioblastoma gene therapy and personalized medicine workflows

Inspirit AI Research Internship

Sep 2022 - Mar 2023

Machine Learning Research Intern

- Built **skin cancer detection pipeline** using OpenCV preprocessing and feature extraction fed into scikit-learn classifiers (Decision Tree, KNN) achieving **93% classification accuracy** via hyperparameter tuning and cross-validation; published as **peer-reviewed paper** through Inspirit AI
- Separately trained a Random Forest classifier on miRNA sequences achieving **95% predictive accuracy ($p < 0.05$)** for cancer cell likelihood prediction, validated via stratified k-fold cross-validation across multiple data splits

TECHNICAL PROJECTS

Lotus Health (<https://github.com/AkashGogate/lotus-health>)

March 2026

- Financial risk modeling engine on **1,080-node ICD-10 comorbidity graph** integrating MEPS and CMS data across **45M insurance records**; RAG pipeline with semantic similarity search; full React frontend with Cytoscape.js interactive disease graph and voice interface

Self-Improving LLM Agent (<https://github.com/AkashGogate/SelfImprovingLLMAgent>)

Jan 2026 - May 2026

- Architected failure-driven self-improvement loop across a **5,000+ task benchmark**, outperforming ReAct by **15pp**; built custom benchmark infrastructure (AgentBench, terminal-bench, 6-task deterministic suite) with LLM-as-judge harness and matplotlib evaluation pipeline

LLM-Based SAT Practice Test Generator (<https://github.com/AkashGogate/SATPracticeTestGenerator>)

Jun 2024

- Built full-stack app (TypeScript/React.js + Node.js/FastAPI); integrated GPT-4o for adaptive question generation via clean RESTful API with session state, answer validation, configurable difficulty, and multi-question test session support

USTA Tournament Explorer (<https://github.com/AkashGogate/myUSTA>)

Jun 2022

- Built Android app consuming USTA's GraphQL API to surface local tournament data via RESTful API integration; sole developer end-to-end

Ecological Conservation Game (Runescape Parody) (<https://github.com/AkashGogate/EcologicalConservationBots>)

Jan 2024

- Engineered multi-threaded, AI-driven multi-agent simulation in C++ with Raylib; autonomous bots compete under ecological constraints with emergent behaviors; real-time game loop, event system, and collision detection

LEADERSHIP

Princeton Racket Club

May 2024 - Aug 2024

Tennis Coach and Tournament Director

- Directed tournament logistics; coached athletes across skill levels in cross-functional environment; managed end-to-end client communications

Tennis Racket Stringing Services

Jan 2019 - Present

- Founded and scaled to **50+ clients**; manage all operations, inventory, and client consultations